



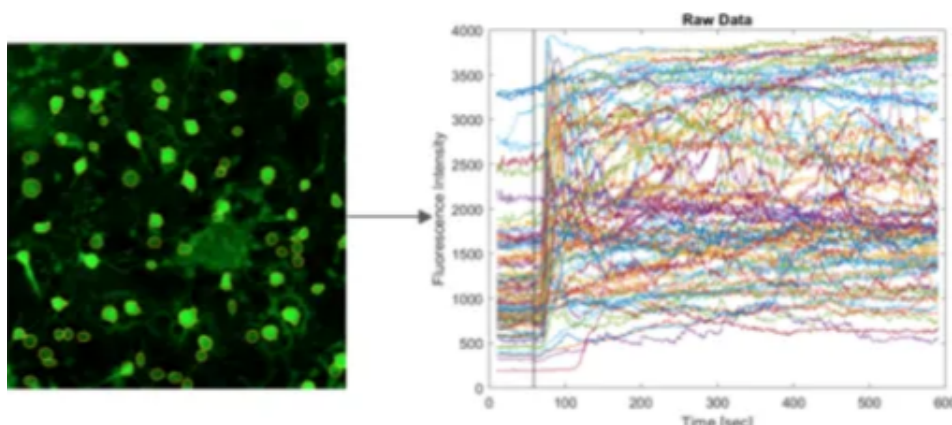
Signal Processing Hub

This section covers the second stage of the OPCal pipeline: processing calcium signals extracted from the ROIs detected during the image analysis step.

Add relevant scripts

Input: Raw Calcium Traces

Raw intensity signals (**F**) extracted per ROI from the cell detection stage.



Each signal represents the fluorescence intensity of a single ROI over time. The input format is a **Time × ROI matrix** (rows = frames, columns = ROIs).

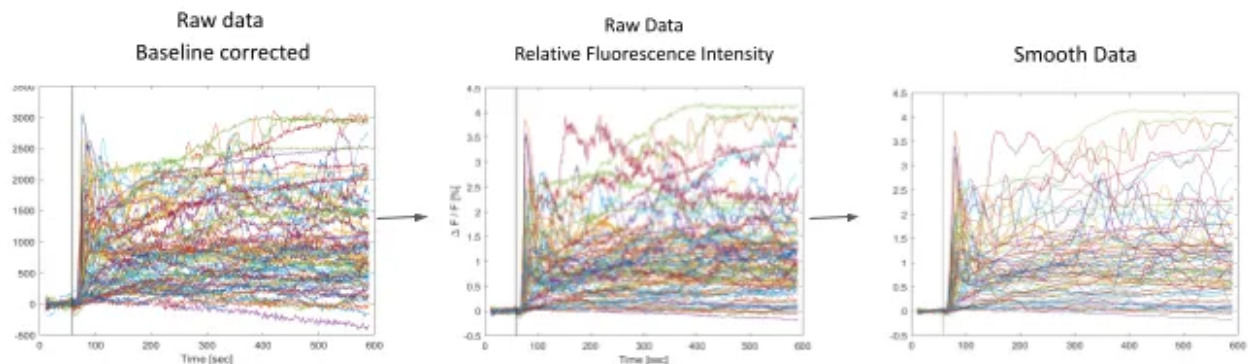
Example:

Add an example for time X intensity matrix

Benchmark data - [download here](#)

Pre-processing

Pre-processing is a critical step to normalize calcium signals and reduce noise before further analysis. In this step, we transform raw intensity data into $\Delta F/F_0$, smooth the traces, and optionally detect peaks for analysis and training our classification model.



Pre-processing steps:

- **Trimming:** Remove the first 10 seconds (unstable baseline)
- **Baseline Correction:** Subtract the mean fluorescence of the baseline period. There are several approaches to baseline correction. **MAD?**

Baseline is computed by pre-stim median:

It takes the window before the stim was given and calculates the median in it.

Advantages:

- median over mean - less sensitive to noises as mean calculation
- provides a more stable baseline

In addition, multiple baseline correction approaches are offered for the user's analysis.

- **Normalization ($\Delta F/F_0$):** $\Delta F/F_0 = (F - F_0)/F_0$
 - both options are presented to user, subtract and $\Delta F/F_0$
- **Smoothing:** Apply Gaussian or Savitzky–Golay filters.
 - Gaussian:

- Pros: Strong noise reduction, produces smooth and continuous traces
- Cons: Broadens peaks, reduces amplitude, blurs sharp onsets/offsets
- Savitzky-Golay:
 - Pros: Preserves peak shape, maintains amplitude better than Gaussian.
 - Cons: Less effective at noise suppression if the window is too small, risk of artifacts ("ringing") with large windows

Examples:

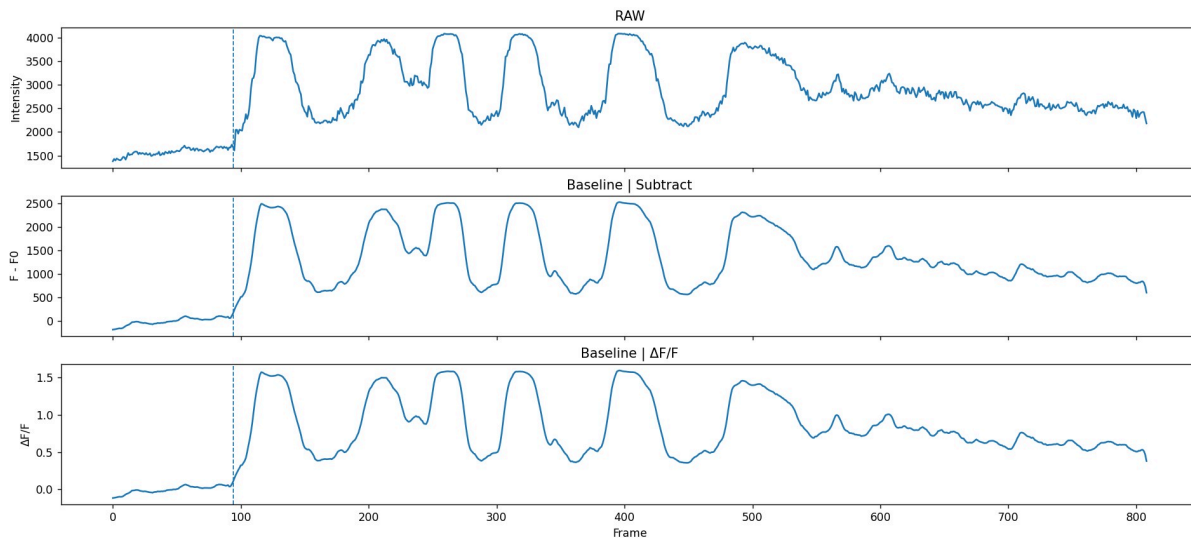
Baseline computed - Pre-stim median

Both normalization methods are presented in addition to the raw data.

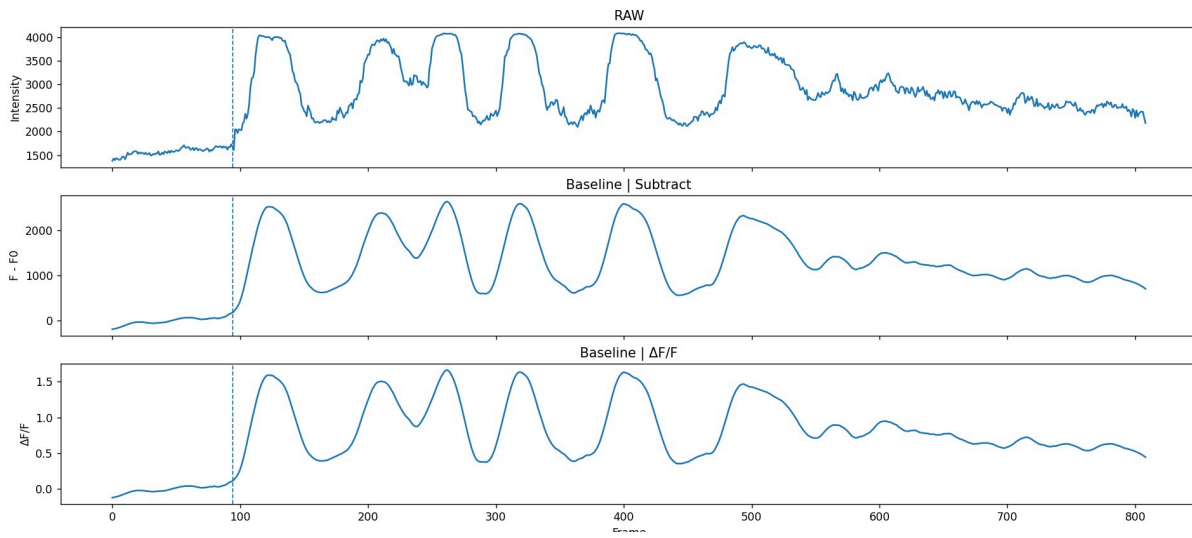
Main differences are between Gaussian and Savitzky-Golay filters.

1. Using Savgol:

window = 10

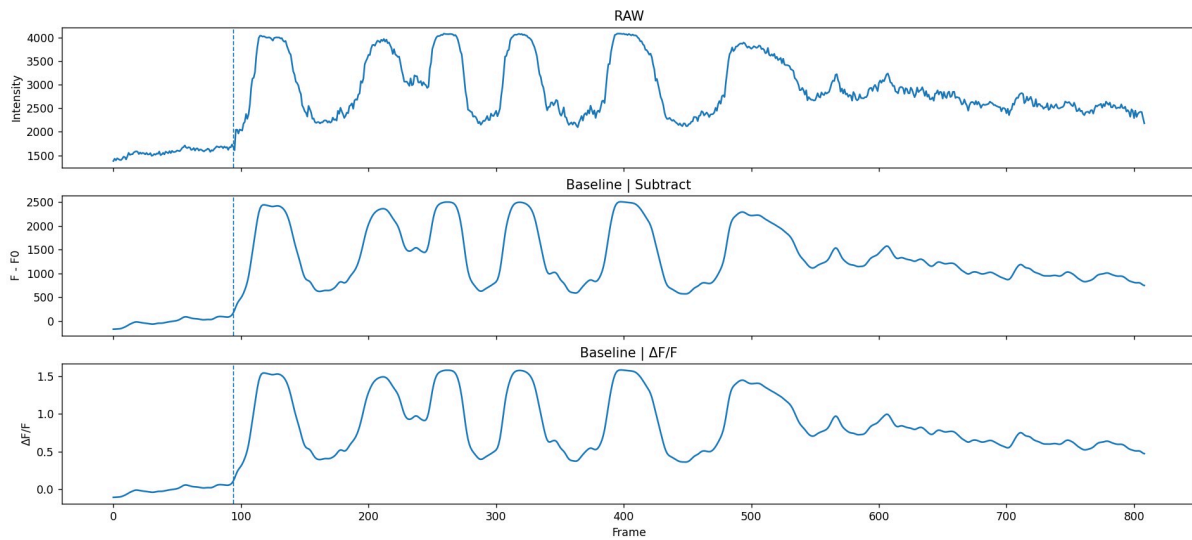


window = 30

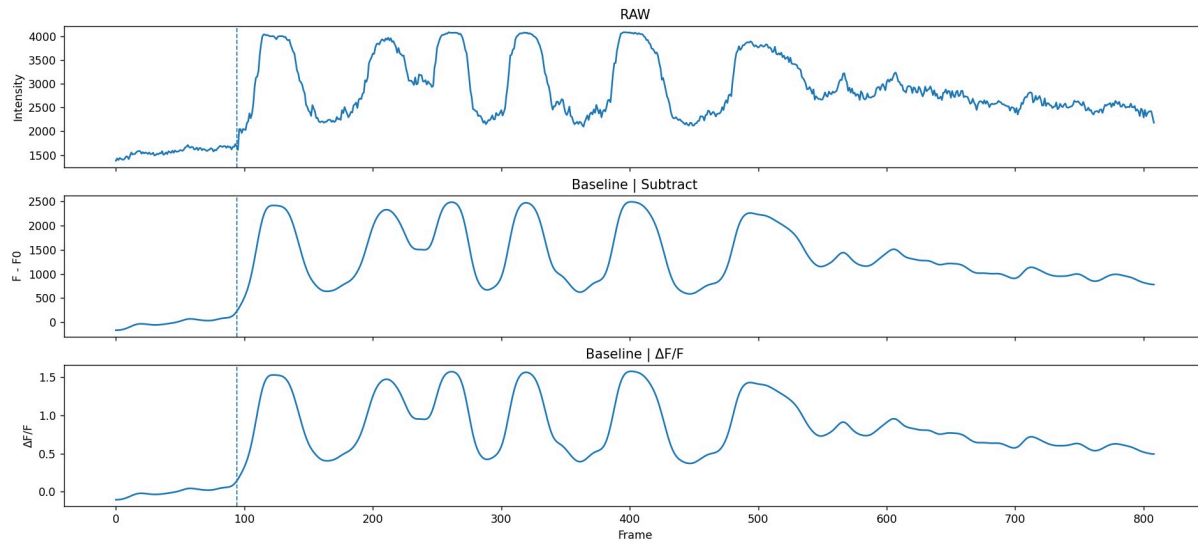


2. Using Gaussian:

$$\sigma = 2$$



$$\sigma = 4$$



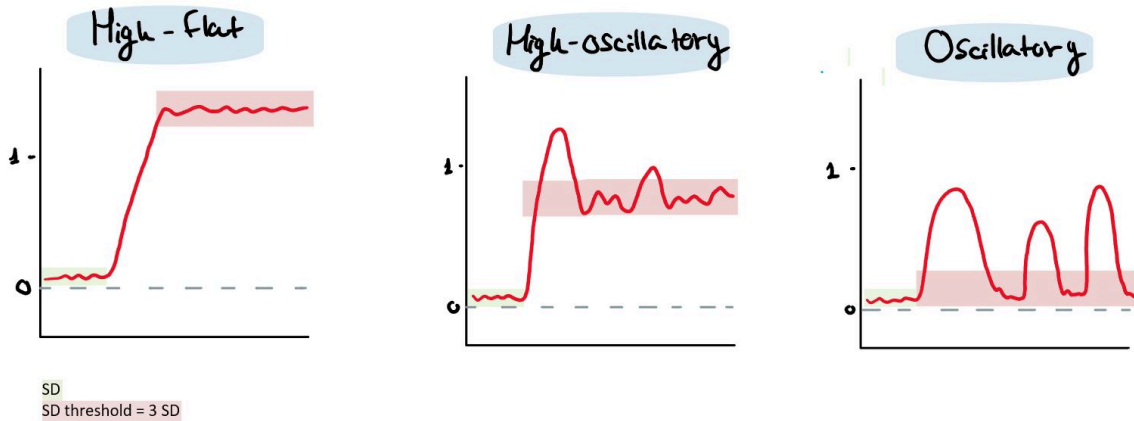
Glossary

Term	Definition
ROI	Region of Interest – individual cell body from segmentation
F	Raw fluorescence intensity
F_0	Baseline fluorescence
$\Delta F/F_0$	Relative change in fluorescence, highlights biological activity

Preprocess signals for classification

To transform raw calcium traces into a format that better reflects true biological activity, highlights signal patterns such as peaks or sustained activity, and supports reliable manual or automated classification.

Standard deviation threshold



Manual Labeling Tool

This stage of the pipeline provides an interactive workspace for manual annotation of calcium traces after signal pre-processing. The tool is organized so the user goes through four steps:

1. **Session setup** – create a new session, resume an existing one, or load directly from a folder.
2. **Upload and indexing** – load the signal data and assign identifiers to detected cells.
3. **Labeling workspace** – the main interface where users inspect traces (raw or smoothed), switch display modes, and assign labels to biological events.

Labeling parameters

Sampling rate (Hz) 1.08

☒ Show raw signal

☒ Show smoothed signal

☒ Apply Savitzky-Golay smoothing

Smoothing window (samples) 31

Polynomial order 3

Baseline method rolling_median

Rolling median window (s) 20

SD threshold k 3.00

Stimulus time (s) 50.00

OPCAL-Labeler

Manual labeling tool • v1.0.0-rc1

1 Start session 2 Upload & indexing 3 Label files 4 Finish & export

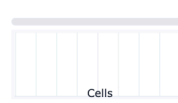
Cells

Session:

/Users/shakedschmarch/OPCAL_LA
BELS/Labeled signals 810
frames_160225 - NEW -
Oscillatory/20250820_133818_000
02

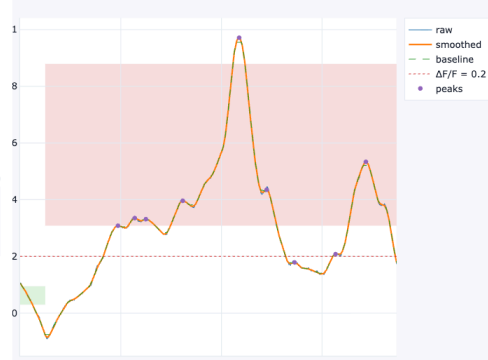
Cell index

0



Progress: 0 / 83 labeled

Cell cell_000000



Label

- Class
- ☐ High-flat
 - ☐ High-oscillatory
 - ☒ Oscillatory
 - ☐ Low-activity
 - ☐ Drifting

☐ Mark as uncertain

Notes

Optional free text

Save label (CSV)

Back

Next

Save at least one label to proceed to Finish & export.

OPCAL-Labeler v1.0.0-rc1 • Local labeling tool • MIT/BSD-style license. No telemetry. Data stays local.

4. **Export** – save annotations and metadata into structured CSV files for downstream analysis and reproducibility.

Session structure

For each session, a dedicated directory is created with the following content:

- **session.csv** – session metadata (recording information, user settings, and parameters).
- **labels.csv** – all manual annotations enriched with processing parameters.
- **peaks.csv** – optional peak events detected or confirmed during labeling.
- **cell_map.csv** – mapping between cell indices and assigned identifiers.

Data export

Annotations are stored in **labels.csv**. Each row represents a single cell annotation and includes both manual labels and the processing parameters active during labeling. The schema includes:

- **session_id**: Unique identifier of the session

- `recording_id` : Recording name or identifier
- `annotator_id` : Identifier of the person labeling
- `saved_utc` : Timestamp of when the label was saved (UTC)
- `cell_index` : Numeric index of the cell (zero-based)
- `cell_id` : Formal cell identifier string
- `label` : The assigned category (e.g. *Drifting*, *Oscillatory*)
- `notes` : Optional free-text notes by the annotator
- `filter_type` , `filter_window` , `filter_polyorder` : Signal smoothing parameters in effect
- `baseline_method` , `baseline_window_s_or_q` : Baseline correction method and window
- `sd_method` , `threshold_k` : Statistical thresholding parameters used
- `mean` , `std` , `rms` : Summary statistics of the trace after processing
- `frac_above_thr` : Fraction of samples above threshold
- `peaks_per_min` : Event frequency if peaks were detected
- `version` : Tool or schema version

This structure ensures that annotations are tied directly to the processing context in which they were made, allowing full reproducibility and consistent downstream use.

Signal classification

Class	Definition	Criteria	Example note
High-flat	Elevated plateau with little/no oscillations	Rise above baseline, sustained plateau, ≤1 peak after rise	One peak then stable high level
Oscillatory	Recurrent calcium events without sustained plateau	≥2 peaks above $\Delta F/F$ threshold, baseline returns between peaks	3 oscillations, baseline returns each time

Class	Definition	Criteria	Example note
High-oscillatory	Elevated plateau with superimposed oscillations	Sustained high level + ≥ 2 peaks on top of plateau	High plateau with 4 peaks on top
Low-activity	No significant activity	Flat trace, no events crossing $\Delta F/F$ threshold	Minor noise only
Low-activity (single-event)	Single transient without sustained plateau	1 peak only , signal returns fully to baseline	One isolated peak
Drifting	Continuous trend without discrete events	Gradual upward/downward slope, no clear plateau or peaks	Bleaching-related decay
Decay	Normal rise or plateau followed by collapse	Sharp rise/oscillations then continuous downward slope	Signal rises then decays fully

- **Drift** = baseline never stable to begin with → often easier to filter out.
- **Decay** = signal looks OK at first → trickier, because it contaminates “good” classes if left unhandled.

Event detection

Feature extraction

Feature dashboards

